GOVERNMENT OF INDIA

Ministry of Earth Sciences

Understanding the Earth.
Securing the Future

Confidential and Proprietory
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SHRI NARENDRA MODI

HONORABLE PRIME MINISTER, INDIA



MESSAGE

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MESSAGE

Scientific understanding of the Earth's systems is essential for sustainable development, climate resilience, disaster preparedness, environmental protection, and the responsible utilization of natural resources. The Ministry of Earth Sciences plays a vital role in advancing research, observation systems, weather forecasting, ocean sciences, seismology, climate studies, and polar research that support informed decision-making across multiple sectors of the economy.

The Chamber of Commerce and Industries of India (CCI India) recognizes that scientific innovation and technological advancement in earth sciences have far-reaching implications for agriculture, infrastructure, energy, transportation, disaster management, maritime industries, and environmental sustainability. Through policy engagement, industry partnerships, research collaboration, innovation promotion, startup support, and investment facilitation, CCI India seeks to strengthen collaboration between scientific institutions and industry.

This Knowledge Paper provides comprehensive insights into the Ministry's strategic priorities, research ecosystem, technological capabilities, policy initiatives, and emerging opportunities in weather services, ocean economy, climate science, marine technologies, geospatial applications, environmental monitoring, and disaster risk reduction. It also highlights opportunities for public-private collaboration, technology commercialization, and innovation-led development.

The publication is intended to benefit policymakers, scientific institutions, researchers, industry leaders, technology companies, startups, investors, educational institutions, and development organizations interested in advancing India's scientific capabilities and environmental resilience. By encouraging collaborative research and innovation, it aims to strengthen evidence-based policymaking, improve service delivery, and support sustainable national development.

As India prepares to address future environmental and climate challenges, science and innovation will remain at the forefront of national progress. CCI India looks forward to supporting the Ministry of Earth Sciences in building a resilient, knowledge-driven, and technologically advanced ecosystem for the benefit of society and the economy.

Munindra Kumar
Director General
CCI India

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PREFACE

India's ambition to become a Viksit Bharat by 2047 rests not merely on economic growth, but on the nation's capacity to understand, manage, and leverage the natural systems that sustain life, enable commerce, and protect communities. The Ministry of Earth Sciences (MoES) stands at the sovereign intersection of scientific knowledge, national security, and sustainable development mandated with the profound responsibility of monitoring and predicting the Earth's atmosphere, oceans, seismic behaviour, climate systems, and polar environments.

In an era defined by accelerating climate disruption, expanding blue economy aspirations, disaster-risk volatility, and the digital transformation of geospatial intelligence, the Ministry of Earth Sciences has quietly evolved from a weather-forecasting body into one of India's most strategically consequential scientific ministries. Its mandate now encompasses monsoon prediction with direct implications for ₹18 lakh crore in agricultural output, tsunami and earthquake early warning systems protecting over 7,500 km of coastline, ocean resource mapping underpinning India's Deep Ocean Mission, polar scientific missions asserting India's global standing, and climate adaptation frameworks that will shape urban planning and infrastructure investment for decades.

Yet, the Ministry operates largely within a government-to-government paradigm, with limited formalised mechanisms for industry engagement, private sector co-creation, startup innovation, investment facilitation, and global knowledge partnerships. This structural gap represents not a failure of the Ministry, but an opportunity for a credible, structured, and mission-aligned industry body to serve as a strategic bridge channelling private capital, sectoral expertise, innovation capacity, and global networks toward the Ministry's national objectives.

This Knowledge Paper has been prepared by the Chamber of Commerce and Industry of India (CCI India) to articulate a comprehensive strategic collaboration framework between CCI India and the Ministry of Earth Sciences. Grounded in rigorous analysis of the Ministry's institutional mandate, budgetary evolution, sectoral landscape, gap assessment, and emerging opportunities, this paper positions CCI India as a multi-dimensional implementation partner serving simultaneously as policy facilitator, investment promotion partner, startup accelerator, knowledge partner, capacity builder, and ecosystem architect.

The paper is intended for submission to and review by Union Ministers, Secretaries to the Government of India, NITI Aayog, senior bureaucrats, industry leaders, development finance institutions, international organisations, and institutional investors. It is designed to initiate a formal, structured, and long-term collaboration that aligns private sector capabilities with India's earth sciences agenda contributing measurably to national resilience, scientific advancement, economic competitiveness, and global leadership.

We present this paper with conviction in the transformative potential of government-industry co-creation, and with the institutional commitment of CCI India to serve as a trusted, capable, and responsible strategic partner to the Ministry of Earth Sciences.

Chamber of Commerce and Industry (CCI India)

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About CCI India

CCI India (The Chamber of Commerce and Industry of India) is a premier national business chamber dedicated to fostering economic growth, industry development, entrepreneurship, innovation, and investment promotion across India. As the voice of Indian business and industry, CCI India serves as a bridge between government, industry, startups, academia, investors, and civil society, facilitating meaningful collaboration for nation-building and sustainable development.

Through policy advocacy, business facilitation, research, stakeholder engagement, capacity building, and strategic partnerships, CCI India works to create an enabling environment for enterprises of all sizes. The Chamber actively supports key national initiatives such as Startup India, Make in India, Digital India, Atmanirbhar Bharat, Skill India, and Viksit Bharat 2047 by connecting innovative enterprises with government opportunities and promoting inclusive economic development.

CCI India operates through sector-specific councils, industry forums, knowledge platforms, and state-level engagements that address emerging opportunities and challenges across sectors including manufacturing, agriculture, technology, healthcare, education, infrastructure, sustainability, MSMEs, and startups. The Chamber regularly organizes policy dialogues, investment summits, business delegations, workshops, conferences, and knowledge-sharing initiatives to strengthen India's economic ecosystem.

With a strong commitment to innovation, entrepreneurship, and public-private collaboration, CCI India strives to empower enterprises, facilitate investment, enhance competitiveness, and contribute to India's journey toward becoming a globally competitive and economically resilient nation. Its role as a knowledge partner and industry catalyst enables ministries, departments, state governments, startups, and businesses to collaborate effectively in achieving national development objectives.

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SECTION 1: INTRODUCTION TO THE MINISTRY OF EARTH SCIENCES

1.1 Constitutional and Institutional Mandate

The Ministry of Earth Sciences (MoES) was formally constituted in 2006 through a restructuring of the Ministry of Ocean Development, consolidating India's earth science institutions under a unified scientific governance framework. Its constitutional relevance derives from the Government of India (Allocation of Business) Rules, 1961, which vest in the Ministry the responsibility for Earth observation, atmospheric sciences, oceanography, seismology, climate science, and polar research functions that directly support the constitutional directive to safeguard the environment and improve public welfare under Article 48A and the broader Directive Principles of State Policy.

The Ministry's institutional mandate covers six interlocking domains: (i) meteorological observation and weather forecasting through the India Meteorological Department (IMD); (ii) oceanographic research and services through the National Institute of Ocean Technology (NIOT) and the Indian National Centre for Ocean Information Services (INCOIS); (iii) seismic monitoring and earthquake preparedness through the National Centre for Seismology (NCS); (iv) climate and atmospheric research through the Indian Institute of Tropical Meteorology (IITM); (v) polar and cryosphere science through the National Centre for Polar and Ocean Research (NCPOR); and (vi) the Deep Ocean Mission, a ₹4,077 crore flagship programme exploring India's exclusive economic zone and beyond.

1.2 Evolution and Governance Role

From its origins as a weather-recording agency in 1875 when the British colonial administration established the India Meteorological Department following the catastrophic Bombay cyclone of 1882 to its present avatar as a multi-domain earth sciences ministry, MoES has undergone a century-and-a-half of institutional evolution. Each phase of national development brought new scientific imperatives: the Green Revolution demanded monsoon precision; nuclear testing required atmospheric modelling; post-2004 Indian Ocean tsunami necessitated real-time ocean surveillance; and the climate crisis demands integrated Earth System Models capable of projecting impacts across sectors.

Today, MoES functions as both a service delivery institution providing weather, ocean, and seismic alerts that protect lives and property and a strategic scientific agency whose research outputs directly inform national planning, food security policy, infrastructure investment, coastal zone management, and India's global climate diplomacy. Its governance role has expanded from reactive monitoring to proactive prediction, and increasingly to scenario modelling that supports policy formulation at the highest levels of government.

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1.3 National Significance

The strategic value of MoES to India's national development is both immediate and existential. In the immediate term, IMD's monsoon forecasts determine irrigation schedules, agricultural input deployment, and commodity procurement decisions affecting over 600 million farmers and agricultural labourers. INCOIS's tsunami early warning system, which demonstrated its efficacy during the 2004 and subsequent seismic events, protects over 250 million coastal residents. In the longer term, the Ministry's climate models and oceanographic data underpin India's Nationally Determined Contributions (NDCs) under the Paris Agreement, its Blue Economy policy, and its Arctic and Antarctic research commitments.

The Ministry also carries significant diplomatic weight. India's meteorological contributions to the World Meteorological Organisation (WMO), its Arctic Council observer status, its engagement with the Intergovernmental Oceanographic Commission (IOC), and the Deep Ocean Mission's submarine vehicle programme (Matsya 6000) collectively position India as a serious global player in Earth system science enhancing the nation's soft power alongside its scientific credibility.

SECTION 2: HISTORICAL TIMELINE OF THE MINISTRY OF EARTH SCIENCES

2.1 Institutional Milestones and Policy Transformation

The institutional history of India's earth sciences apparatus spans 150 years and reflects the nation's evolving relationship with natural systems from colonial-era climate monitoring to 21st-century deep-ocean exploration. The following timeline captures the pivotal milestones that shaped the Ministry's current institutional character.

Year	Milestone	Strategic Significance
1875	IMD Established	Founded by the British colonial government after the 1882 Bombay cyclone, IMD became India's first systematised Earth observation institution, laying the foundation for meteorological governance.
1945	IMD Modernisation	Post-independence, IMD expanded its network of weather stations and introduced synoptic chart analysis, enabling regional weather prediction critical to food security planning.
1981	National Institute of Oceanography (CSIR-NIO)	Establishment of ocean science infrastructure positioned India as a serious maritime research nation, enabling resource mapping of the Indian EEZ and international seabed areas.

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